Padraig Lavery

Forward Deployed Engineer – Full Stack, Cloud & Applied AI

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years delivering software into separate deployments for UK and Middle East banking clients, from front end to back end to the Kubernetes and AWS platform it runs on. Turns messy production problems into permanent fixes, and writes the guides and runbooks that let other teams run what was built. Building applied LLM skills through Incident Copilot, an agentic retrieval-based triage service.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Delivering across client deployments

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals to meet client security and compliance audit requirements.
- Led the technical migration of email templates onto dedicated brand domains across four banking brands (MJML, .NET Core), preserving deliverability.
- Planned testing for the AWS SDK v3 and Node 22 Lambda runtime upgrades across repositories and bank deployments, and authored the engineering migration guide.
- Built full stack features across web and mobile, including a secure e-code resend flow with multi-bank email routing and targeted redemption pop-ups with deep-linking.

Solving production problems

- Resolved a live customer-facing outage in a Kubernetes/Istio ingress setup and permanently corrected the Helm deployment templates.
- Eliminated recurring daily out-of-memory crashes in a high-volume ingestion service by loading caches once per batch instead of per file.
- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation.
- Authored runbooks and SOPs so the production services team resolves recurring issues without engineering escalation.

Working across teams

- Led a cross-team initiative to enable automated API testing for an internal service, from request validation and OpenAPI docs to a stabilised GitHub Actions CI pipeline.
- Built the CI/CD pipeline for a UI Builder service in an internal AI investigation project.
- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost web API using Go, Docker, Kubernetes and Git.

AI PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Takes a production incident description, retrieves relevant runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status. Moved from AWS Bedrock to local models via Ollama to remove inference costs.

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular card transactions, with a model served via SageMaker.

SKILLS

- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL
- **Full stack:** Angular, Node.js, .NET Core, FastAPI, GraphQL, OpenAPI, PostgreSQL, MongoDB
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **AI:** LLM integration, retrieval-augmented generation, agentic tool-calling, Ollama, AWS Bedrock

- **Practices:** production incident response, runbooks and technical guides, security remediation (Snyk, OWASP), mentoring

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026